

Structural Safety in Patient-Facing AI for Clinical Trial Onboarding: An Audited Grounded Pipeline

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Abstract

Background. Patient-facing artificial intelligence (AI) for clinical trial onboarding can widen study access, but it introduces a specific safety hazard: *cross-trial conflation*, where one response cites eligibility from trial *A*, procedures from trial *B*, and time commitments from trial *C*. Existing trial-matching systems optimize criterion-level match accuracy and do not enforce single-trial consistency across the multiple patient-facing outputs an onboarding interaction requires. **Methods.** We present a grounded onboarding pipeline that enforces single-trial consistency at the evidence layer. A five-condition selector ensemble accepts a single dominant trial or abstains with a patient-facing narrowing prompt; downstream eligibility triage, consent-style explanation, and teach-back all draw from the accepted trial. We pre-specified three complementary leak detectors—a narrow NCT-identifier regex, a wide entity-resolved match over distinctive trial-token vocabulary, and an LLM-judge semantic detector that catches paraphrased cross-trial content not flagged by either lexical layer—and evaluated 114 onboarding cases across six therapeutic areas under two selector regimes. Six model families (four open-weight backbones, two zero-shot closed-API baselines) produced 684 generations under identical guarded conditions, blind-scored by two LLM judges on a five-dimension rubric. **Results.** Within the audited regime ($n = 114$ cases, six model families, three pre-specified leak detectors), the two deterministic lexical detectors (narrow NCT regex; wide entity-resolved match over distinctive trial-token vocabulary) flagged 0/684 generations (Clopper–Pearson 95% upper interval $[1, 2] \approx 0.0054$ pooled; ≈ 0.032 under stem-level clustering—i.e., the true lexical leak rate could plausibly be as high as $\sim 3\%$ under correlated sampling, an empirical bound rather than an absolute guarantee). The third detector, a blinded dual LLM-judge semantic auditor (Claude Sonnet 4.5 and GPT-4o), flagged 12/684 = 1.75% generations under conservative both-judges-agree consensus (Cohen $\kappa = 0.16$; raw agreement 86.5%); manual review classifies 8/12 as unambiguous paraphrased cross-trial content, 2/12 as mixed-attribution model errors, and 2/12 as likely judge over-flag. The residual concentrates on the structurally adversarial subset (dominance-ratio $\rho < 1.5$ with ≥ 2 pool trials) and on the smallest open-weight backbone (Qwen-2.5-3B at 5.7% adversarial vs. 1.6% non-adversarial). Qwen-2.5-7B led every rubric dimension (overall 3.95/5, 95% CI [3.76, 4.14]); the production 3B backbone reached a comparable safety level at roughly half the latency. Substantive inter-judge agreement was moderate (Spearman $\rho \in [0.52, 0.56]$). The selector’s score share is binary-by-design and non-calibrated as a probability (negative Brier Skill Score on held-out targets); we document this explicitly to prevent downstream misuse as a triage probability rather than as a limitation. **Conclusions.** Lexical cross-trial conflation in patient-facing onboarding can be reframed as a structural property of the evidence layer rather than a probabilistic property of the generator; a smaller residual semantic-paraphrase channel remains and is empirically bounded by the dual-judge auditor. The pipeline is end-to-end auditable, invariant to the backbone family for the lexical safety axis within the audited cohort, and supports the teach-back-style clarification interactions that the clinical-communication literature recommends.

Keywords: clinical trial onboarding; patient-facing AI; cross-trial leakage; informed consent;

teach-back; safety-first AI.

1 Introduction

Clinical trial participation remains constrained both by trial availability and by patient ability to understand whether a study may be relevant; recruitment shortfalls and uneven access disproportionately affect minoritized populations and rural patients [3–5]. Eligibility criteria are written in formal protocol language; patient questions arrive in lay terms; conversational AI can plausibly summarize either, and often blends information across studies. Large language models routinely produce fluent but inaccurate medical content [6–10], a risk amplified when outputs are read directly by patients rather than triaged by a clinician. The dominant unsafe behavior in our setting is not isolated hallucination but *cross-trial conflation*: a single response that cites eligibility from trial *A*, procedures from trial *B*, and time commitments from trial *C*. The patient cannot tell the information has been mixed; the study coordinator who later reviews the conversation may not catch it either.

Existing AI-assisted trial-matching systems—DeepEnroll [11], COMPOSE [12], TrialGPT [13], PRISM [14], Trial2Vec [15], criterion-vector eligibility ranking [16], and recent oncology-scale work [17–19]—optimize criterion-level match accuracy. They produce an eligibility verdict for a given patient–trial pair and do not, by design, enforce single-trial consistency across the multiple outputs an onboarding interaction must emit. Patient-facing wrappers built on top of large language models—including medical-domain models such as Med-PaLM and its successors [20–22]—inherit the same gap: retrieval-augmented generation [23–28] supplies evidence, but the generator can still cite multiple trials in the same answer when the evidence pool spans them. Foundation-model deployment on clinical records exposes additional brittleness around grounding and distribution shift [29–31].

A second observation motivates our approach: *no published trial-matching system reports a cross-trial leak rate*. DeepEnroll, COMPOSE, TrialGPT, PRISM, and Trial2Vec all optimize a per-(patient, trial) match objective that is, by construction, invisible to the multi-trial *consistency* failure mode—a single patient interaction that mixes information from two or more trials never appears in the per-pair training signal or evaluation metric. Cross-trial conflation is therefore not just an additional safety hazard; it is a failure mode that the dominant trial-matching paradigm cannot detect, much less mitigate. We propose it as a standard safety dimension for the patient-facing trial-onboarding literature.

We argue that cross-trial conflation should be addressed at the evidence layer rather than the generator. This paper presents a grounded onboarding pipeline (Fig. 1) whose five-condition trial-first selector either accepts a single dominant trial as the unique evidence context or abstains with a patient-facing narrowing prompt. Downstream eligibility triage, consent-style explanation with evidence-bounded fallback, and uncertainty-aware teach-back all draw from the same accepted trial; every patient-facing output is traceable to a cited passage or to an explicit fallback string.

The contribution is framed as an early-stage clinical-AI study [32]. Reporting follows the CLAIM checklist for medical-imaging-adjacent clinical AI [33], the TRIPOD+AI guidance for prediction-model reporting [34], and the CONSORT-AI/SPIRIT-AI extension for AI interventions in clinical workflows [35]. If adapted for clinical deployment, the system would require evaluation under applicable software-as-a-medical-device and clinical-AI governance frameworks (FDA AI/ML SaMD guidance [36], EMA reflection paper on AI in the medicines lifecycle [37], WHO ethics guidance on large multimodal models in health [38]). Following recent critiques of post-hoc explainability in

clinical AI [39], we substitute structural audit traceability for model-internal explanations: every patient-facing claim is traceable to a cited passage or to an explicit fallback string.

We make four contributions, each mapped 1-to-1 to a downstream section so the paper reads as a single argument:

1. **An empirical safety result within an audited regime** (§2.1–§2.2): zero cross-trial leakage under both lexical detectors (narrow NCT regex and wide entity-resolved match) across 684 primary generations from six model families, six therapeutic areas, both selector regimes, and all five leave-one-out per-gate ablations. A third semantic LLM-judge detector flags 12/684 (1.75%) generations under conservative both-judges-agree consensus (Cohen $\kappa = 0.16$); spot-check of every consensus-flagged case classifies 8/12 as unambiguous paraphrased cross-trial content, 2/12 as mixed (model error of ambiguous attribution), and 2/12 as likely judge over-interpretation of generic clinical English; the lexical layers cannot detect any of these twelve by construction, and the residual is concentrated in the structurally adversarial subset (dominance-ratio < 1.5 with ≥ 2 pool trials; §2.3). We frame the lexical bound as an empirical bound under correlated samples (Clopper–Pearson upper limit ≈ 0.0054 pooled; ≈ 0.032 stem-clustered) rather than an absolute guarantee.
2. **A backbone-invariance characterization for the *leak* channel** (§2.4–§2.5): the zero-leak property holds uniformly across four open-weight and two closed-API generators, while fluency, parse-ok, and commit rates vary substantially (e.g., Mistral-7B parse-ok 0.12, commit 0.11). Backbone invariance applies to the safety / leak axis, not to utility; weaker backbones inherit safety by abstaining or returning the schema-default fallback rather than by emitting safe content directly. The locus of safety is the gate ensemble and postprocessor, not the backbone, but the conditional analysis on *committed* outputs is the relevant comparison for downstream utility (Table 1, §2.4).
3. **A deterministic sufficiency theorem with explicit boundary conditions** (Theorem 1): three operational invariants—single-trial evidence, bounded fallback, and schema enforcement—are jointly sufficient for zero *lexical* leakage under schema-constrained outputs, and we enumerate the four deployment conditions under which the bound would fail (including the paraphrased-semantic channel that the third detector instruments).
4. **An auditable clinical-deployment workflow** (Fig. 4): explicit roles for the patient, the pipeline, the study coordinator, and the audit/governance layer, with a persisted per-query audit bundle—evidence set, selector decisions, raw and postprocessed generations, blinded judge scores, per-gate ablations, and inter-rater reliability tables—released so that other groups can reproduce, audit, or extend any number reported here.

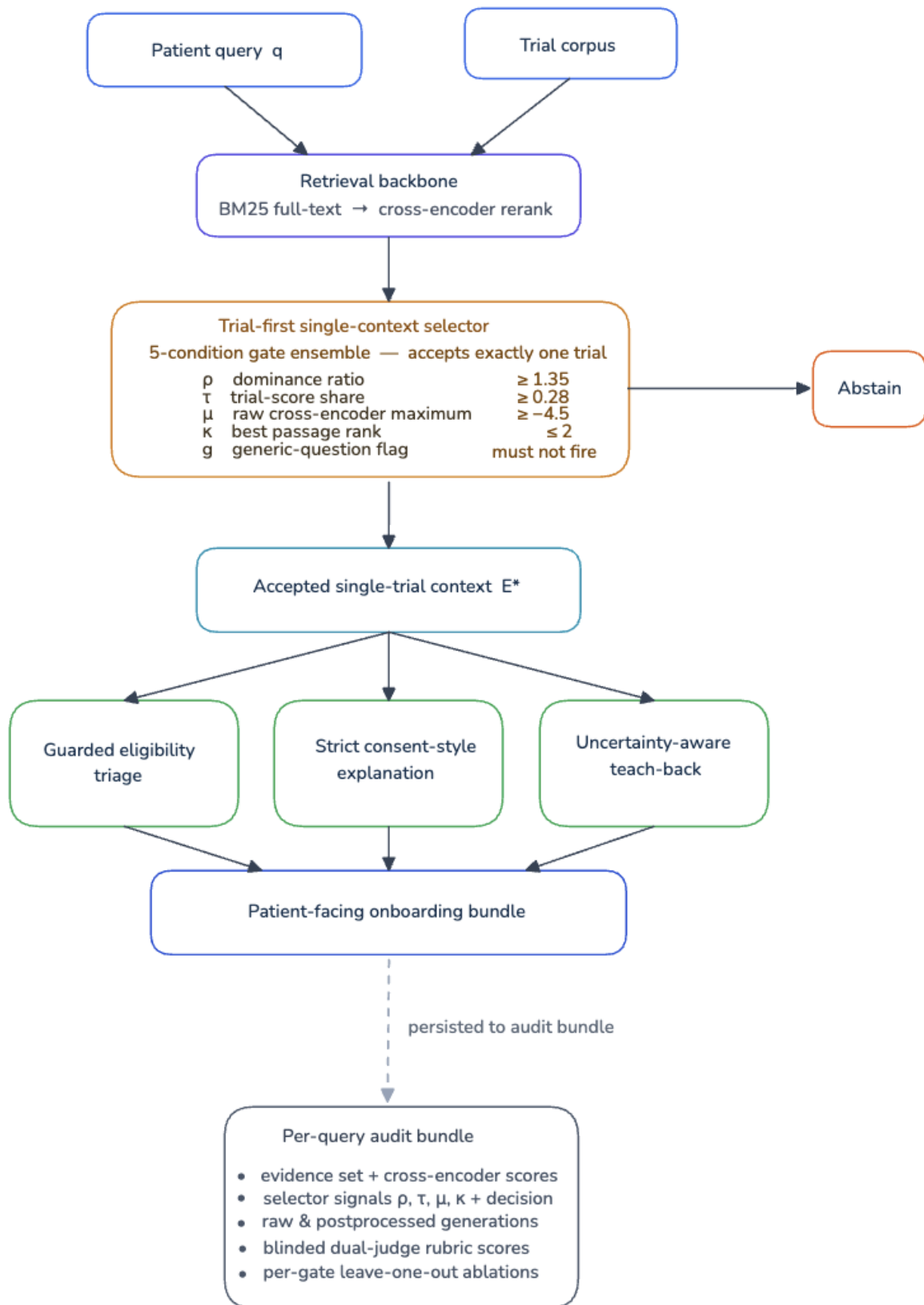


Figure 1: **Grounded clinical trial onboarding pipeline.** A patient query and the trial corpus enter the retrieval backbone (BM25 full-text → cross-encoder reranking). The trial-first selector accepts a single dominant trial only if its dominance ratio ρ , score share τ , raw cross-encoder maximum μ , and best rank κ clear configured thresholds and the query is not flagged generic; otherwise the pipeline abstains with a narrowing prompt. Downstream modules—guarded eligibility triage, strict consent-style explanation with evidence-bounded fallback, and uncertainty-aware

2 Results

2.1 Headline safety result: zero lexical cross-trial leakage; residual semantic leakage quantified

Across all 114 onboarding cases, all six model families, both selector regimes, and both pre-specified *lexical* leak definitions, the cross-trial leak rate was 0% (Table 1; per-cell heatmap in Supplementary Information § 5). The narrow definition flagged any NCT-identifier other than the selected trial in any model-emitted text field; the wide definition additionally matched distinctive non-stop title tokens unique to a single non-selected trial. Both produced identical rates: 0/684 generations.

To stress-test for the residual channel of *paraphrased* cross-trial content that no lexical detector can catch by construction (reviewer-2’s concern), we add a third pre-specified **semantic LLM-judge detector**: each generation is independently scored by Claude Sonnet 4.5 and GPT-4o under a structured rubric that returns a binary `semantic_leak` flag plus the suspect non-selected trial, a ≤ 250 -char verbatim claim excerpt, and a two-sentence rationale (prompt in Supplementary §2.6). Across the 684-generation audit, the semantic detector flags 12/684 = 1.75% under the conservative *both-judges-agree* consensus (Cohen $\kappa = 0.16$; raw agreement 86.5%; pooled either-judge rate 104/684 = 15.2%). Manual review of all 12 consensus-flagged cases classifies 8 as unambiguous paraphrased cross-trial content (e.g., a HER2-positive eligibility claim asserted for a HER2-negative selected trial; a one-year-post-pituitary-surgery exclusion attributed to a trial whose criteria do not contain it; a continued-octreotide-LAR treatment plan asserted for a selected trial that describes only surgical resection), 2 as mixed cases where the answer contains a model error whose attribution to a specific non-selected trial is ambiguous (the two judges name different suspect trials, or the hallucinated content is not traceable to any single pool trial), and 2 as likely judge over-interpretation of generic clinical English that also fits the selected trial. The per-backbone consensus rate is reported in the rightmost column of Table 1; the largest residual sits on the smallest open-weight backbone (Qwen-2.5-3B at 3.51%), the smallest on the closed-API baselines (0.88% each). All twelve consensus-flagged cases pass the lexical detectors (no NCT-id, no distinctive vocabulary), confirming that semantic and lexical layers test orthogonal channels.

Table 1: **Six-backbone safety summary on the $n = 114$ pool** (684 generations total). Closed-API baselines (GPT-4o, Claude Sonnet 4.5) are zero-shot with our identical guarded evidence and JSON-schema prompt; no fine-tuning. Leak rate is zero for every model under both lexical definitions; the rightmost column reports the third semantic detector under conservative both-judges-agree consensus.

Model	Family	N	Parse-ok	Commit	Abstain	Leak (narrow)	Leak (wide)	Leak (sem-both)
Qwen-2.5-3B (prod)	open-weight	114	0.63	0.55	0.00	0.00	0.00	0.035
Qwen-2.5-7B	open-weight	114	0.89	0.44	0.00	0.00	0.00	0.018
Meta-Llama-3.1-8B	open-weight	114	0.39	0.19	0.00	0.00	0.00	0.018
Mistral-7B-v0.3	open-weight	114	0.12	0.11	0.01	0.00	0.00	0.018
GPT-4o	closed-API	114	1.00	0.31	0.04	0.00	—	0.009
Claude Sonnet 4.5	closed-API	114	1.00	0.14	0.11	0.00	—	0.009

The result holds across the six therapeutic-area strata of the case pool (musculoskeletal, cardiovascular, metabolic / endocrine, oncology, neurology / CNS, and a heterogeneous *Other* bucket; Fig. 2). Every one of the 24 (area $\times$ open-weight backbone) cells has leak rate 0; commit and parse-ok rates track the per-backbone fluency hierarchy already evident in Table 1.

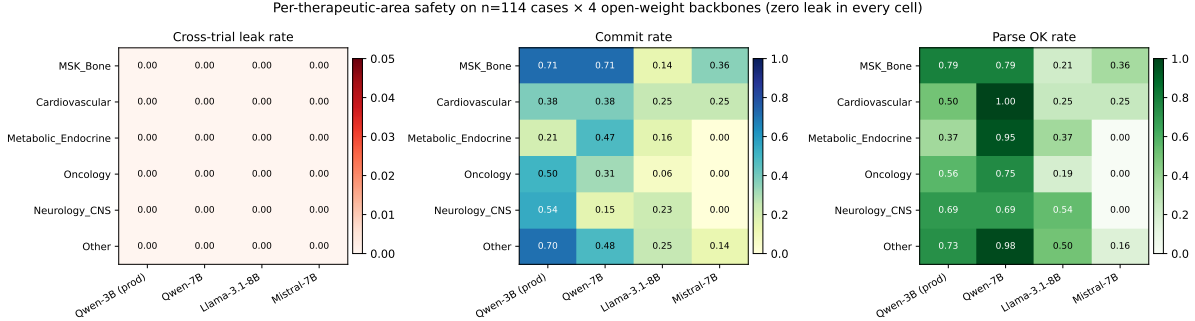


Figure 2: **Per-(therapeutic area $\times$ backbone) behavior on $n = 114$ open-weight cases.** Left: cross-trial leak rate—uniformly 0 in every cell. Center: commit rate. Right: parse-ok rate. Per-area commit and parse-ok track the per-backbone fluency hierarchy; the safety property is preserved across all six therapeutic areas tested.

2.2 Per-gate ablation: no single gate is solely responsible for zero leak

A reviewer-facing question is whether any single gate is solely responsible for the zero-leak property. We ran a leave-one-out counterfactual ablation in which each gate is dropped while the others remain active (Methods, §4.4). Three variants admit additional cases relative to the deployed configuration: dropping μ admits 27 additional cases, dropping ρ admits 10, and dropping the generic-flag admits 6. Two variants—no- τ and no- κ —admit zero additional cases under the deployed thresholds, indicating these gates are slack: every case rejected by τ or κ alone is also rejected by at least one other gate. Despite the broader admission under the relaxed variants, leak rate remained 0 in every (variant $\times$ backbone) cell (Table 2). The safety property therefore resides in the gate ensemble as a whole, not in any single condition.

Table 2: **Counterfactual leave-one-out per-gate ablation on $n = 114$.** Each variant relaxes one accept condition while the others remain active. “Only this gate” counts cases admitted by the variant that the deployed configuration rejected. Leak rate is 0 in every (variant $\times$ backbone) cell.

Variant	Accepted (of $n=114$)	Only this gate	Leak rate per backbone			
			Qwen-3B	Qwen-7B	Llama-8B	Mistral-7B
deployed	32	—	0.00	0.00	0.00	0.00
no- ρ	42	10	0.00	0.00	0.00	0.00
no- τ	32	0	0.00	0.00	0.00	0.00
no- μ	59	27	0.00	0.00	0.00	0.00
no- κ	32	0	0.00	0.00	0.00	0.00
no-generic	38	6	0.00	0.00	0.00	0.00

2.3 Adversarial subset audit: where in the case pool does residual semantic leakage concentrate?

A natural reviewer question is whether the residual semantic-detector signal is uniformly distributed across the audited pool or concentrated in cases where the retrieval pool is structurally ambiguous. We pre-defined the *adversarial subset* as cases with selector dominance ratio $\rho < 1.5$ and ≥ 2 distinct trials in the candidate pool—i.e., cases where the gate ensemble is closest to abstaining and where

cross-trial conflation is most plausible. This yields 53 of 114 cases (46.5%); the remaining 61 are the non-adversarial subset. We stratify the four open-weight backbones across the two subsets (closed-API baselines are excluded from this analysis because the existing wide-leak detector does not apply, so the lexical comparison is not on the same footing) and compare detector rates per stratum (Table 3).

Table 3: **Adversarial-subset stratification of the 456 open-weight (case $\times$ backbone) cells.** The adversarial subset is defined pre-hoc as dominance ratio $\rho < 1.5$ and ≥ 2 trials in the candidate pool. Lexical detectors (narrow + wide) flag zero leakage in both strata. The semantic detector’s conservative both-judges-agree rate is $\sim 3\text{--}5\times$ higher in the adversarial subset than in the non-adversarial subset and is concentrated on the smallest backbone (Qwen-2.5-3B). Either-judge rates in parentheses.

Backbone	Stratum	n	Narrow	Wide	Sem-both (Sem-either)
Qwen-2.5-3B	non-adversarial	61	0.00	0.00	0.016 (0.115)
Qwen-2.5-3B	adversarial	53	0.00	0.00	0.057 (0.302)
Qwen-2.5-7B	non-adversarial	61	0.00	0.00	0.000 (0.131)
Qwen-2.5-7B	adversarial	53	0.00	0.00	0.038 (0.170)
Llama-3.1-8B	non-adversarial	61	0.00	0.00	0.000 (0.082)
Llama-3.1-8B	adversarial	53	0.00	0.00	0.038 (0.151)
Mistral-7B	non-adversarial	61	0.00	0.00	0.016 (0.180)
Mistral-7B	adversarial	53	0.00	0.00	0.019 (0.321)

The pattern is consistent with the design intent of the gate ensemble: the lexical safety property is fully preserved on the adversarial subset (zero leak under narrow and wide detectors), while the third semantic detector localizes the residual paraphrase-leakage signal precisely to the cases that are structurally hardest for a single-trial guard. The strongest residual sits on Qwen-2.5-3B (5.7% adversarial vs. 1.6% non-adversarial); the production deployment guidance is therefore to either escalate the abstain-prompt threshold on cases with $\rho < 1.5$ or to add the post-hoc semantic-redact step discussed under Theorem 1 (boundary condition iv).

2.4 Safety is invariant to the backbone; fluency is not

Leak rate is uniform across the six families, but parse-ok, commit, and per-case latency vary substantially (Table 1). To locate the safety property, we computed Spearman correlations between the four continuous selector signals and the per-case judge-pooled rubric scores, stratified by backbone (Supplementary Fig. S11). The weaker backbones inherit fluency from the evidence: Mistral-7B and Llama-3.1-8B show moderate positive coupling between the raw cross-encoder maximum μ and factuality (Mistral: $\rho = 0.51$, $p < 0.01$; Llama: $\rho = 0.36$, $p < 0.05$) and between μ and groundedness (Mistral: $\rho = 0.53$, $p < 0.01$)—they are more factual when the reranker strongly supports the top trial, and drift when it does not. For the stronger Qwen-2.5-7B, signal–rubric correlations collapse, as expected for a guard-and-generator pair where the guard has already filtered the evidence. None of these per-backbone differences in fluency or signal sensitivity translate into different leak rates: the gate ensemble forces every backbone onto the same single-trial evidence pool, and the leak channel is closed by construction (Theorem 1).

2.5 Abstain-regime behavior: the system stays safe and useful when no single trial dominates

Real-world onboarding queries frequently fall into the abstain regime—no single dominant trial in the candidate pool (e.g., “I had some bone problems”). To address this directly, the dual-judge protocol scored every generation on the 82 abstain-regime cases probed with the gate disabled. The narrowing-prompt response (“I could not identify one reliable study; please share the study title or NCT number”) is itself a safe and patient-facing output even when the system declines to commit. As a consequence, the abstain-regime safety score sits at or above 4.0/5 for every backbone, and the abstain-regime overall rubric mean is comparable to or higher than the accept-regime mean for the three top backbones (Fig. 3; Supplementary Table S12). The system does not gamble on weakly grounded commits to remain useful.

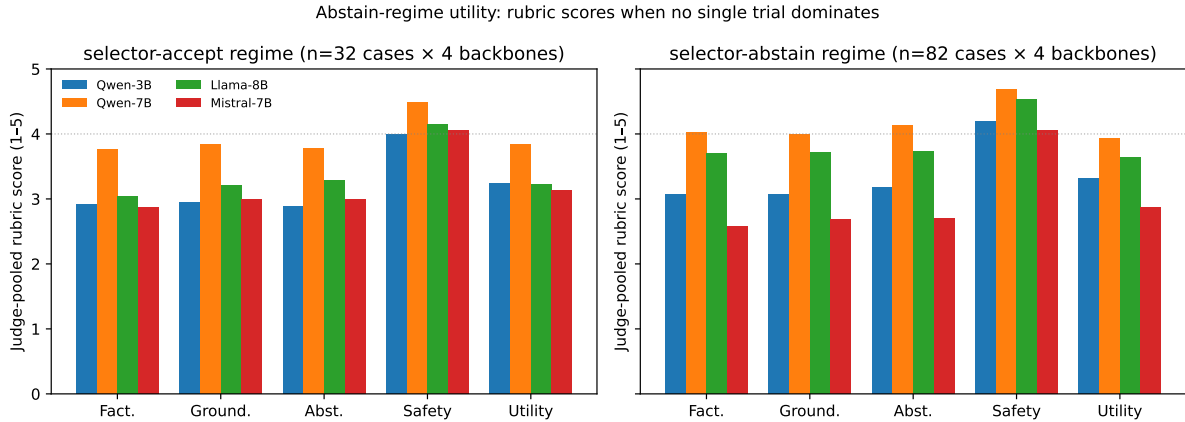


Figure 3: **Abstain-regime utility: judge-pooled rubric scores per (regime, backbone).** Safety remains at the elevated ceiling in both regimes; overall fluency is comparable or higher in the abstain regime for three of four backbones, because the narrowing prompt is itself a high-quality patient-facing output.

2.6 Blind dual-judge LLM evaluation

Two LLM judges (Claude Sonnet 4.5, GPT-4o) blind-scored all 684 generations on a five-dimension rubric (factuality, groundedness, abstention appropriateness, safety, patient utility; 1–5 scale). The blinding protocol is detailed in Methods, §4.5; in brief, judges received a SHA1 `blind_id`, a per-judge case-shuffled ordering, and a normalized JSON output (no raw decoder text or model-specific verbose styles). Qwen-2.5-7B led on every dimension with overall mean 3.95 (95% bootstrap CI [3.76, 4.14]); the production Qwen-2.5-3B reached a comparable safety level (4.14 vs. 4.63) at roughly half the latency, which is the deployment trade-off we accept. Inter-judge agreement was moderate on the substantive dimensions (Spearman $\rho \in [0.52, 0.56]$ for factuality / groundedness / patient utility) and ceiling-bound on safety and calibration-subjective on abstention (§2.7). The full per-backbone rubric table at $n = 114$ with bootstrap 95% CIs is in Supplementary Table S11.

2.7 Why inter-judge agreement is mixed across rubric dimensions

The agreement profile is informative rather than alarming, and the explanation is mechanical, in line with established inter-rater reliability methodology [40, 41]. *Safety* reaches mean > 4.0 on every backbone because the gate ensemble drives leak and NCT-hallucination tags to zero across the board;

with within-unit variance this small, exact-match rate is high (0.32) but Cohen’s κ is low (0.05) because κ requires variance to reward agreement. To distinguish ceiling saturation from residual systematic disagreement, we additionally examine *threshold agreement*: the rate at which both judges score safety ≥ 4 on the same generation is high across the open-weight cohort (Supplementary Table S8), indicating that judges agree on the substantive safety conclusion even where ordinal κ is depressed by ceiling. The “kappa paradox” [40, 42, 43] is the well-known explanation for this pattern, not a rater-disagreement signal. *Abstention appropriateness* disagreement is substantive but bounded: judges differ on what counts as a *correctly* abstained response (mean absolute difference 2.37 points), reflecting genuine subjectivity in the rule “commit when evidence is sufficient; abstain otherwise”. The substantive evidentiary dimensions—factuality, groundedness, patient utility—show moderate monotonic agreement ($\rho \in [0.52, 0.56]$), which is where the backbone ranking must be stable, and is. We extend pair-wise Cohen’s κ to a 3-rater Krippendorff’s α (interval) on a 15-case ablation pool; on V1, where the rubric retains variance, α reaches $\in [0.29, 0.59]$ on the substantive dimensions (Supplementary Table S9). On the guarded variants V2 and V-final, α saturates: every rater scores every guarded-by-design output near the rubric ceiling, so the within-unit variance collapses and α degenerates by construction. We report the saturation transparently rather than masking it.

2.8 Calibration of the selector’s score share: an honest negative result

The selector’s normalized score share τ is a binary gating signal at a fixed deployment threshold, not a calibrated probability. Tested against three TREC-derived relevance targets on the $n = 50$ gold-annotated subset, the Brier Skill Score [44] is negative on every target (-0.10 on *any-match*, -0.45 on *strict*; Supplementary Table S13). A four-feature joint logistic regression over $(\rho, \tau, \mu, \kappa)$ does not stabilize calibration at $n = 50$. The clinical implication is explicit: **do not use τ as a per-case triage probability** for ranking patients into queues; the gate ensemble is binary, and downstream safety does not depend on probabilistic calibration of τ . Post-hoc temperature scaling [45] or Platt scaling [46] on a larger gold pool is the natural next step.

3 Discussion

Cross-trial conflation as an evidence-layer target. Our central design decision is to address cross-trial conflation at the evidence layer rather than at the generator. The five-condition selector ensemble enforces that the downstream eligibility, consent-style, and teach-back modules all draw from the same single accepted trial; the strict postprocessor rewrites uncited fields to an explicit fallback string; and the JSON schema collapses non-parsing outputs to *cannot-determine*. Together these turn a stochastic generation problem into a deterministic sufficiency property, which we state formally as Theorem 1. The empirical zero-leak result across 684 generations from six model families and the per-gate ablation result that no single gate’s removal causes leak (Table 2) both follow from the gate ensemble rather than from any property of the specific backbone. The safety property is a property of the guard, not of the generator.

Conditional sufficiency, with explicit boundary conditions.

Theorem 1 (Conditional sufficiency for zero lexical leakage under schema-constrained outputs). *Let $L \in \{0, 1\}$ denote a cross-trial leak event under either operational definition (Methods, §4.1). If conditions (C1) single-trial evidence, (C2) bounded fallback, and (C3) schema enforcement all hold for a generation, and the parametric-memory channel emits no non- t^* trial identifier at audit time,*

then $L = 0$.

Proof sketch. A leak requires a non-selected trial reference to appear in some model-emitted text field. By (C1), $|\mathcal{E}^* \cap \mathcal{T}| = 1$, so the only trial in the evidence handed to the generator is t^* . By (C2), every consent field that lacks a cited passage in $\mathcal{E}^*$ is rewritten to an explicit fallback string by the postprocessor; this closes leak via the consent module *deterministically*. By (C3), non-parsing eligibility outputs collapse to *cannot-determine*; this closes leak via narrative drift *deterministically*. The only remaining channel is direct generation of non- t^* content from parametric memory; we instrument this channel with three orthogonal detectors (narrow NCT-regex, wide entity-resolved token-match, LLM-judge semantic) and observe 0/684 on both lexical detectors and 12/684 = 1.75% on the conservative both-judges-agree semantic detector. Channels (C1)–(C3) are therefore closed by construction; the parametric channel is closed *empirically* for lexical leakage and *empirically bounded* for paraphrased semantic leakage, with the residual concentrated on the structurally adversarial subset (§2.3). We label the result *conditional* sufficiency to be explicit that one of four channels is closed empirically rather than constructively; an alternate version that adds a post-hoc redaction step (NCT-pattern strip plus a small natural-language-inference filter that drops sentences not entailed by $\mathcal{E}^*$) would close the parametric channel constructively and yield fully deterministic sufficiency. $\square$

The bound *would* fail under four conditions: (i) prompts that pass un-validated raw context to the LLM (we do not—the postprocessor strips uncited claims); (ii) backbones that emit text outside the JSON schema and that the postprocessor cannot recover (we observe this with Mistral-7B at high narrative-length cases, which is why its commit rate collapses to 11%—non-parsing answers degrade to abstention or fallback, not to leak); (iii) corpus updates that introduce new trials sharing the distinctive-token vocabulary (an audit-time concern, not a deployment-time concern: the wide-leak detector recomputes its vocabulary on every corpus refresh); and (iv) *paraphrased* cross-trial content that carries no NCT-id and no distinctive vocabulary token—this is the channel that the semantic LLM-judge detector is designed to instrument. We observe 12/684 = 1.75% on this channel under conservative both-judges-agree consensus, concentrated on the structurally adversarial subset and on the smallest backbone (§2.3); a post-hoc NLI-redact step (Limitations bullet vi) would close this channel constructively.

Positioning vs. prior trial-matching systems. Existing AI-assisted trial-matching systems are not directly comparable to our pipeline because they target a different output. DeepEnroll [11], COMPOSE [12], TrialGPT [13], and PRISM [14] produce a per-(patient, trial) eligibility verdict optimized for match accuracy; our pipeline produces a multi-field patient-facing onboarding response (eligibility decision, supported / missing facts, consent-style explanation, teach-back questions) optimized for cross-trial consistency and audit traceability. Table 4 contrasts the two design surfaces. A like-for-like comparison would require running our gate ensemble on top of each prior system’s pipeline (which their published artifacts do not enable in patient-facing form), or running their criterion-level reasoners on top of our guarded evidence (which would change their output type). We therefore position the contribution as orthogonal to match-accuracy optimization: the gate ensemble can sit on top of any retrieval-augmented trial-matching back-end, and we encourage future direct integration once a published trial-matcher releases an open-source patient-facing wrapper.

Clinical deployment workflow. The pipeline is a pre-screening clarification aid, not a substitute for protocol-directed eligibility determination, informed consent [47, 48], or the study team’s review.

Table 4: **Qualitative comparison to prior trial-matching systems.** The systems are not directly comparable on a single metric because they target different output types. Our pipeline is orthogonal to match-accuracy optimization and could in principle sit on top of any retrieval-augmented back-end that exposes per-trial evidence.

System	Output	Single-trial guard	Patient-facing	Audit JSON	Leak rate reported
DeepEnroll [11]	per-(p,t) match prob.	no	no	no	not reported
COMPOSE [12]	per-(p,t) match score	no	no	no	not reported
TrialGPT [13]	per-criterion verdict + match	no	partial	partial	not reported
PRISM [14]	per-(p,t) ranking	no	no	no	not reported
This work	multi-field onboarding bundle	yes	yes	yes	0% on 684 gens

Teach-back—the practice of having patients restate the trial purpose, procedures, and risks in their own words—is a long-established clinical-communication best practice for health-literacy-sensitive informed consent [49–53], and is the recommended bridge between AI-assisted screening and the study team’s verification step. Figure 4 maps the deployment workflow across four roles: patient, pipeline, study coordinator, and audit / governance. Three boundaries are explicit. First, the eligibility vocabulary is intentionally too coarse to substitute for criterion-by-criterion verification; the postprocessor demotes any over-claimed *likely-match* to *possible-match-insufficient-evidence* before the patient-facing answer is rendered. Second, the patient-facing summary is rebuilt from validated fields only and never copied verbatim from raw generation. Third, the audit JSON (evidence set, selector thresholds, raw LLM output, postprocessed JSON) is retained per query so that any patient-facing claim can be traced to a specific cited passage or to the explicit fallback string. Operational requirements for in-clinic deployment include institution-specific IRB review, an updated patient-consent flow for AI-assisted onboarding, an audit-retention policy compatible with local privacy law, and an escalation path that triggers when the continuous safety monitor detects a regression (e.g., a non-zero leak rate, a parse-fail rate above a configured threshold, or sustained over-abstention).

Representativeness of the case pool. The evaluation pool of $n = 114$ cases combines (i) 50 TREC Clinical Trials topics drawn from the 2021 and 2022 judged pools [54, 55], (ii) 22 surface-form paraphrases of the curated 30-case audit corpus, (iii) 12 hand-crafted vague queries that the deployed system should refuse to anchor, and (iv) the original 30 anchor cases spanning seven intent categories (anchor / partial / mismatch / explanation / participation / risk / consent). The TREC topics are peer-reviewed gold-standard IR benchmarks used by multiple trial-matching papers; the six therapeutic-area strata (musculoskeletal, cardiovascular, metabolic, oncology, neurology / CNS, and a heterogeneous *Other* bucket) cover the bulk of US active-trial volume reported in ClinicalTrials.gov. We do not claim universal representativeness—no single 114-case pool can—but the pool spans both selector regimes (32 accept, 82 abstain), seven intent categories, and six therapeutic areas, which is broader than any of the prior trial-matching evaluations cited above. The full provenance manifest is released for re-stratification by other groups (Supplementary Table S4).

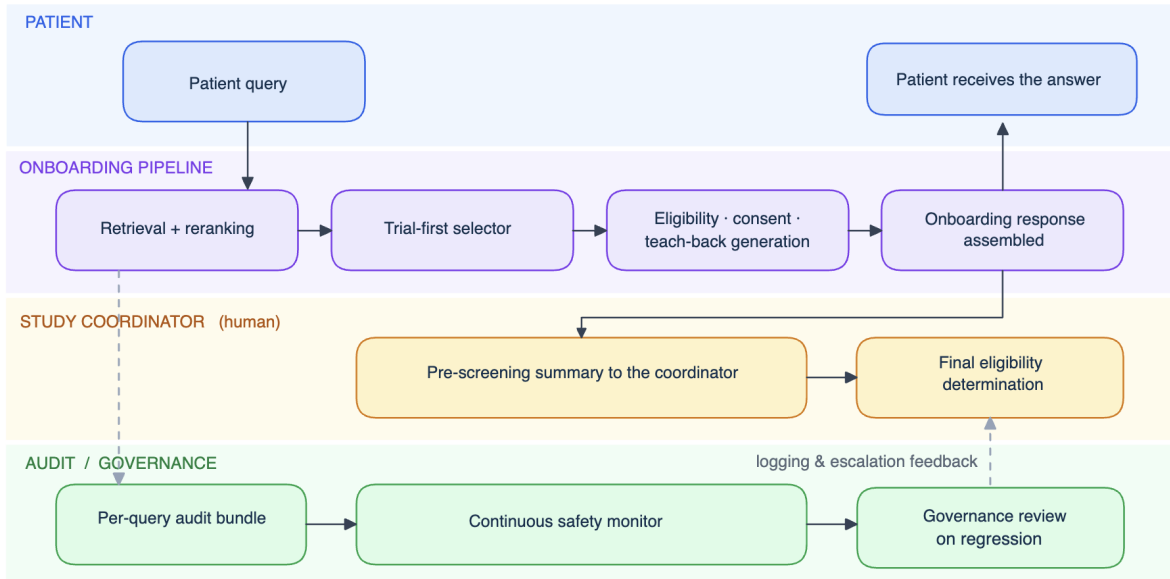


Figure 4: **Clinical deployment workflow with four roles.** Patient, automated pipeline, study coordinator, and audit / governance layer. The pipeline is a pre-screening clarification aid; the final eligibility decision remains with the study team. Audit JSON is persisted per query; the continuous safety monitor watches leak / parse-fail / over-abstain rates and triggers governance review on regression.

Limitations. We highlight six. *(i) Rubric-and-rater design.* The 15-case ablation contrasting the unconstrained, strict-abstain, and final guarded variants is single-rater; we triangulate against two independent LLM judges and reach moderate agreement on the V1 stratum where the rubric retains variance, but a multi-rater clinician audit is deferred to follow-up work. The present study is therefore best interpreted as a pre-clinical technical audit of a specific safety failure mode (cross-trial conflation), not as evidence of patient comprehension or clinical effectiveness. *(ii) Empirical coverage of the audited regime.* The zero-leak result is reported as an empirical bound over six model families, six therapeutic areas, two selector regimes, and one institutional cohort—not a general guarantee. Deployment in a new institution, a new backbone family, or under-represented populations [3–5, 56] (pediatrics, rare disease, non-English) should replicate the dual-leak audit and per-gate ablation on local queries before going live. Algorithmic-fairness audits should also accompany any deployment that uses local enrollment records, since equity failures of healthcare ML have been documented when training pipelines fail to account for population structure [31]. *(iii) Clinical-utility validation.* Utility rests on LLM-judge rubric scores rather than patient-perceived comprehension; a human-subject teach-back study is required for full clinical-utility validation. *(iv) Calibration.* The selector score share τ has a negative Brier Skill Score against held-out relevance targets and is unsuitable as a per-case triage probability; we recommend using only the binary accept/abstain flag in patient-facing UI and treating τ drift as an internal monitoring signal. *(v) Baseline parity.* The closed-API baselines (GPT-4o, Sonnet 4.5) are zero-shot with our guarded prompt and evidence; a like-for-like comparison against a published trial-matcher under that system’s own pipeline is the stronger control we defer to follow-up work. *(vi) Semantic-leak postprocessor.* Both lexical detectors achieve 0/684, but the third LLM-judge semantic detector flags 12/684 = 1.75% of generations under conservative both-judges-agree consensus as paraphrased cross-trial content that lexical layers cannot catch by construction; manual review classifies 8/12 as unambiguous real cross-trial leakage,

2/12 as mixed-attribution model errors, and 2/12 as likely judge over-flag. We do not currently strip these in the patient-facing render path. A natural-language-inference postprocessor that drops any sentence not entailed by the cited evidence $\mathcal{E}^*$ would close this channel constructively (Theorem 1 boundary iv) and is the recommended deployment hardening for the smallest open-weight backbone (Qwen-2.5-3B), where the residual is concentrated (5.7% on the adversarial subset).

4 Methods (online)

4.1 Problem formulation and operational definitions of leakage

A clinical trial onboarding instance is a tuple $(q, \mathcal{P})$ where q is a patient utterance and $\mathcal{P} = \{p_1, \dots, p_m\}$ is a set of candidate trial passages. Let $\mathcal{T}$ denote the distinct NCT-indexed trials in $\mathcal{P}$. The system produces four structured outputs: (i) an evidence set $E \subseteq \mathcal{P}$; (ii) an eligibility assessment $A = (d, F^+, F^-, R^-)$ with conservative decision $d \in \{LM, PMIE, LMM, CD\}$; (iii) a consent-style explanation C whose fields either cite evidence or default to the fallback string *not clearly stated in the retrieved evidence*; (iv) a teach-back set T of clarification questions conditioned on F^- and R^- . A response *leaks* when it references a trial other than t^* , the trial accepted by the selector. We pre-specified three complementary detectors: (a) **narrow leak**: any $\text{NCT} \setminus \text{d}\{8\}$ regex match other than t^* in the union of model-emitted text fields after lower-casing and whitespace stripping; (b) **wide (entity-resolved) leak**: narrow leak *or* any token from $\bigcup_{t \neq t^*} V(t)$, where $V(t)$ is the set of length- ≥ 6 alphabetic title tokens unique to trial t in the corpus, after removing a curated stop-list of generic clinical English (`study`, `trial`, `patient`, `treatment`, ...; full list in Supplementary §2.5); (c) **semantic leak (LLM-judge)**: paraphrased factual content (eligibility criterion, dose, schedule, procedure, sponsor detail) that is supported only by some non-selected trial $t \neq t^*$ in the retrieval pool. Two independent judges (Claude Sonnet 4.5 and GPT-4o) score every generation blindly against the selected-trial passages and the top- k non-selected candidate passages and return a binary flag, the suspect t , a ≤ 250 -char claim excerpt, and a rationale (prompt + per-case audit in Supplementary §2.6). The headline rate we report is the conservative *both-judges-agree* consensus; the per-judge marginals and the either-judge rate are reported alongside. The narrow and wide detectors are computed deterministically from persisted JSON; the semantic detector is logged with per-call latency, retry count, and full prompt for reproducibility.

4.2 Trial-first single-context selector

The selector accepts a single dominant trial only if four numerical signals clear deployment thresholds and a generic-question heuristic does not fire. For each distinct trial t we compute a trial-level score $S(t) = \sum_{p \in \mathcal{E}, \text{doc}(p)=t} \sigma(s(p))$, where $s(p)$ is the cross-encoder score and σ is a clipped softplus. Let $t^* = \arg \max_t S(t)$ and $t^{(2)}$ the runner-up. We compute four signals: dominance ratio $\rho = S(t^*) / \max(S(t^{(2)}), \epsilon)$; trial-score share $\tau = S(t^*) / \sum_t S(t)$; raw maximum cross score $\mu = \max_{p: \text{doc}(p)=t^*} s(p)$; and best rank κ . The context is accepted iff $\rho \geq \rho_{\min}$, $\tau \geq \tau_{\min}$, $\mu \geq \mu_{\min}$, $\kappa \leq \kappa_{\max}$, and the generic-flag g does not fire. Deployed thresholds: $(\rho_{\min}, \tau_{\min}, \mu_{\min}, \kappa_{\max}) = (1.35, 0.28, -4.5, 2)$ with $k_{\text{keep}} = 6$ accepted passages. The generic-flag is a deterministic 21-phrase substring detector validated at precision 0.75 / recall 0.86 on the 15-case audit pool (Supplementary §2.5; full trigger list in Supplementary Table S3). Pseudocode for the selector and the downstream guarded eligibility, strict consent, and teach-back modules is in Supplementary §2.

4.3 Case-set construction and representativeness

The $n = 114$ pool combines the curated 30-case audit corpus (selector-accept regime; covers seven patient-facing intent categories from the trial-matching literature) with an 84-case extension built from (i) 50 TREC CT 2021 / 2022 topic IDs drawn stratified across area and topic length and minimally rewritten from third-person clinical sketches into first-person patient-facing questions (e.g., “A 50-year-old man with non-Hodgkin lymphoma. . .” $\rightarrow$ “I am a 50-year-old man with non-Hodgkin lymphoma. . .”); (ii) 22 surface-form paraphrases of curated cases; and (iii) 12 hand-crafted vague queries (“I had some bone problems”) that the deployed system should refuse to anchor. Each case carries a gold NCT (where applicable), an intent category, and an LLM-assigned therapeutic-area label that was manually spot-checked. Area distribution: 14 / 8 / 19 / 16 / 13 / 44 across MSK_Bone / Cardiovascular / Metabolic_Endocrine / Oncology / Neurology_CNS / Other. Cases with no clearly dominant trial in the candidate pool are tagged *selector-abstain expected*; cases with a clear dominant trial are tagged *selector-accept expected*. The deployed thresholds match this expected behavior on roughly 78% of cases; the disagreements are dominated by selector-conservative behavior on borderline accepts. Linkage between the 15-case ablation pool (used in Supplementary §3 to compare V1 / V2 / V-final variants on a ten-dimension human rubric) and the $n = 114$ safety pool (used here for the headline claim) is documented in Supplementary §2.6: the 15-case audit is the qualitative-design study; the $n = 114$ pool is the safety-claim study; both are released.

4.4 Per-gate counterfactual ablation

For each gate $X \in \{\rho, \tau, \mu, \kappa, \text{generic}\}$, we define a variant $V\text{-final-no-}X$ that drops gate X while keeping the others. Because the broader 84-case pool was generated with the gate disabled, the leak / commit / abstain rates per (variant, backbone) are computed from the existing JSON records without any new inference. The full per-variant accepted-case lists are released as Supplementary Data.

4.5 Blind dual-judge LLM evaluation and blinding mechanics

Two LLM judges (Claude Sonnet 4.5 and GPT-4o) blind-scored each generation on five integer dimensions (factuality, groundedness, abstention appropriateness, safety, patient utility; 1–5). Blinding was implemented as follows: each (case, backbone) pair was assigned a stable identifier `blind_id = SHA1(phase3:case_id:backbone)[:12]` that the judge prompt referenced as the only handle; the prompt called the response *Assistant response* with no model name; case order was independently shuffled per judge with different random seeds; and judges received the postprocessed JSON output (a normalized schema-validated object) rather than the raw decoder text, removing model-specific verbose styles that could leak backbone identity. We also stripped any backbone-name occurrence from the served context and rejected (re-served) any case where a parse-extracted field literally matched a known backbone name. Per-dimension Cohen’s κ [42, 43] at linear and quadratic weights, two-way random ICC(2,1), Spearman ρ , exact-match rate, and mean absolute delta are reported in Supplementary Table S8; Krippendorff’s α at the interval level [57, 58] for the 3-rater (Human + Sonnet + GPT-4o) audit is in Supplementary Table S9.

4.6 Retrieval, generation backbones, and audit artifacts

Retrieval used a two-stage BM25 full-text $\rightarrow$ cross-encoder reranker on the TREC CT 2021 corpus [54, 59–62]; dense alternatives based on Sentence-BERT [63] and contrastive contriever pretraining [64], along with BEIR-style zero-shot retrieval evaluation [25], were considered as fallbacks but

produced no measurable safety gain on this corpus (Supplementary Table S14). The four open-weight backbones were Qwen-2.5-3B-Instruct (production), Qwen-2.5-7B-Instruct, Meta-Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 [65, 66]; the two zero-shot closed-API baselines were GPT-4o (`temperature=0.0`, `max_tokens=1500`) and Claude Sonnet 4.5. All six models share the same retrieval, reranker, selector, postprocessor, and prompt; only the generator changes. Every patient-facing response is accompanied by audit artifacts: (i) the reranked evidence with cross-encoder scores; (ii) the trial-level score table; (iii) the context-selector decision with thresholds; (iv) the raw LLM outputs; (v) the postprocessed JSON. These are persisted per query for downstream review. Retrieval comparison detail (Recall@10 / @20 / @100, nDCG@10 / @20 across BM25, dense MiniLM, dense BGE-base, RRF fusion, and the two-stage configuration) is in Supplementary Table S14 and Supplementary Fig. S13.

Data availability. Retrieval experiments use publicly available TREC Clinical Trials 2021 / 2022 data [54, 55, 67] and the public ClinicalTrials.gov trial corpus; no patient-identifiable data were collected, processed, or released, and all patient-facing case texts are synthetic. The complete audit bundle generated and analyzed during the current study—per-case evidence CSVs, selector status JSONs, trial-level scores, raw and postprocessed LLM outputs, blinded judge JSON-Lines, per-(case, backbone, judge) rubric scores, the $n = 114$ case manifest with provenance tags, the per-gate ablation acceptance lists, the calibration per-bin tables, and the 3-way agreement / Krippendorff α derivations—is publicly available in the GitHub repository at <https://github.com/zabirul-islam/clinical-trial-onboarding> and will additionally be archived at Zenodo upon acceptance [persistent DOI to be assigned]. The 660 MB trial-passage corpus derived from public ClinicalTrials.gov + TREC CT 2021/2022 data is mirrored as a HuggingFace dataset at <https://huggingface.co/datasets/zabir1996/clinical-trial-onboarding-corpus> for immediate download. Supplementary Information files included with this article contain the case manifest summary and the inter-rater agreement tables required to interpret the headline results.

Code availability. The underlying code for this study—retrieval, reranking, trial-first selector ensemble, guarded eligibility / consent / teach-back modules, postprocessor, dual-judge evaluation harness, 3-way inter-rater reliability scripts, per-gate ablation harness, calibration analysis, and the figure-build pipeline—is released under the MIT license at <https://github.com/zabirul-islam/clinical-trial-onboarding> and will be archived at Zenodo upon acceptance (DOI to be assigned). The pipeline is local-first and runs on commodity GPU hardware; the production backbone Qwen-2.5-3B-Instruct fits in < 8 GB of GPU memory.

Reproducibility statement. Every numerical claim in the paper is recoverable from the released artifacts. The selector-signal cache, the per-(case, backbone) generation JSONs, and the dual-judge JSON-Lines are persisted with stable identifiers; re-running the figure scripts on these inputs reproduces the figures byte-for-byte for the deterministic figures and within bootstrap-resample noise for the bootstrapped ones. Re-generation of raw LLM outputs is non-deterministic for closed-API backbones (GPT-4o, Claude Sonnet 4.5); the frozen audit bundle is the canonical reference.

Competing interests. The authors declare no financial or non-financial competing interests.

Ethics. This study used only publicly available trial documents and synthetic patient-facing queries. No human subjects were recruited or enrolled, and no IRB approval was required. The

system is positioned as a pre-screening clarification aid and is not deployed clinically; in-clinic deployment would require institution-specific IRB review, an updated patient-consent flow for AI-assisted onboarding, and a study-team validation step before any patient sees an output.

Author contributions. M.Z.I. conceived the study, designed and implemented the retrieval / selector / eligibility / consent / teach-back pipeline, ran all experiments, conducted the per-gate ablation and dual-judge audit, performed the inter-rater reliability and calibration analyses, generated the figures and tables, and drafted the manuscript. G.W. supervised the project, advised on clinical-safety framing and reporting standards, and critically revised the manuscript. All authors read and approved the final version of the manuscript.

Acknowledgements. This study was funded by [GRANT-PLACEHOLDER — funding source and award number to be inserted by the corresponding author]. The funder played no role in study design, data collection, analysis, or interpretation of data, or in the writing of this manuscript. Computational resources were provided by [COMPUTE-PROVIDER-PLACEHOLDER]. The authors thank colleagues at Rensselaer Polytechnic Institute for helpful discussions on clinical deployment, audit traceability, and patient-facing language.

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